

OVERHEAD POWER LINES TOOLBOX TALK

Electrical contact is not always required; high voltage can arc across an air gap.

PURPOSE AND SCOPE

Prevent cranes, MEWPs, scaffolds, pipes, ladders and other conductive equipment from approaching overhead electrical lines dangerously.

KEY HAZARDS

1  Crane or MEWP boom approaching an	2  Long metal pipes, ladders or sections	3  Electrical arcing through equipment to ground.	4  Incorrect assumption that rubber tyres
5  Operator blind spots causing incorrect	6  Wet weather increasing electrical exposure.	7  Unknown line height or voltage.	8  Step-potential exposure around energised

CRITICAL POINTS TO VERIFY IN THE FIELD

01 Crane or MEWP boom approaching an energised	02 Long metal pipes, ladders or sections contacting overhead	03 Electrical arcing through equipment to ground.
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REMEMBER — SERNEM

- | | | |
|-------------------------|----------------------------|----------------------------------|
| S Stop and think | E Evaluate the risk | R Remove or reduce |
| N Never assume | E Ensure protection | M Monitor and communicate |

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FIELD REQUIREMENTS AND VERIFICATION

Prevent cranes, MEWPs, scaffolds, pipes, ladders and other conductive equipment from approaching overhead electrical lines dangerously.

SERNEM FIELD 01
FIELD VERIFICATION 1
Line identified

SERNEM FIELD 02
FIELD VERIFICATION 2
Voltage confirmed

SERNEM FIELD 03
FIELD VERIFICATION 3
Clearance defined

SERNEM FIELD 04
FIELD VERIFICATION 4
Spotter assigned

VERIFICATION BEFORE START

- | | |
|--|---|
| ☐ Line identified | ☐ Voltage confirmed |
| ☐ Clearance defined | ☐ Spotter assigned |
| ☐ Barrier installed | ☐ Boom limit checked |

SUPERVISOR DISCUSSION POINTS

- Do we know the line voltage?
- What is our minimum approach distance?
- Who is the dedicated spotter?

FIELD APPLICATION



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CRITICAL CONTROL MEASURES

1. ELIMINATE / REDUCE THE RISK

Confirm line voltage and required minimum approach distance. • Set boom or movement limiting devices where available. • Brief the operator to remain in the cab if equipment contacts a live line unless immediate danger requires evacuation.

2. ENGINEERING & PHYSICAL CONTROLS

Where possible arrange de-energisation and isolation before work. • Do not carry long conductive materials upright beneath overhead lines. • Reassess clearances whenever equipment position or work method changes.

3. WORK METHOD & FIELD CONTROLS

Install physical height restriction goalposts when isolation is not possible. • Clearly mark the approach boundary.

4. PPE, VERIFICATION & SUPERVISION

Use a dedicated spotter for crane and MEWP movements. • Provide sufficient lighting for night operations.

CRITICAL WARNING



Team, high voltage can arc without direct contact. We will treat the line as live, maintain the approved clearance and never move lifting or access equipment close to it without a spotter.

SITE CHECKLIST — BEFORE STARTING WORK

- | | |
|---|---|
| ☐ Line identified | ☐ Voltage confirmed |
| ☐ Clearance defined | ☐ Spotter assigned |
| ☐ Barrier installed | ☐ Boom limit checked |
| ☐ Material route safe | ☐ Lighting adequate |
| ☐ Emergency response briefed | ☐ Crew informed |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			